

OG-Core Examples: TFP, Productivity, Mortality Shocks

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Examples

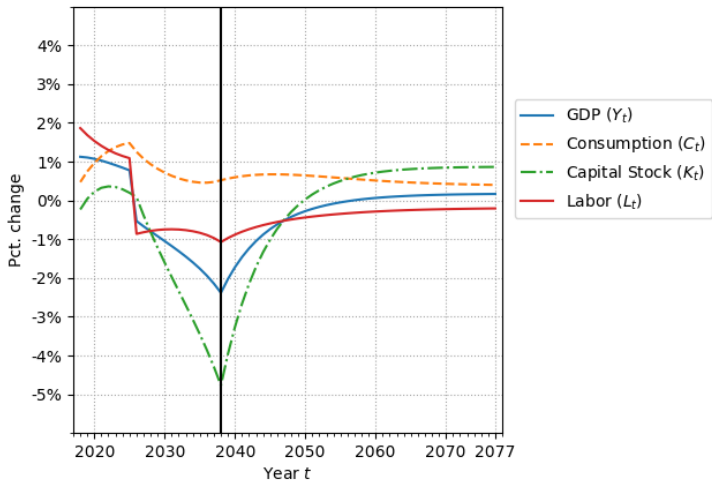
- Tax Cuts and Jobs Act (TCJA) Dynamic Analysis
- Revenue and Macroeconomic Effects of a 70% Marginal Tax Rate
- A Detailed Macroeconomic Analysis of President Biden's 2020 Campaign Tax Proposals
- The Macroeconomics of Improving Biological Aging

TCJA Dynamic Analysis

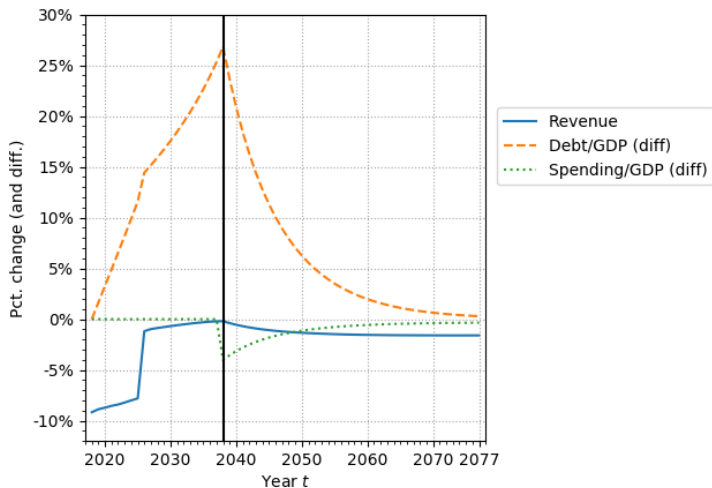
DeBacker, Jason and Richard W. Evans, “[Dynamic Analysis of Tax Cuts and Jobs Act](#),” *Quantitative Notes*, 2018-1, (February 2, 2018).

- GDP growth in first 8 years (1-2% increase)
- Significant crowding out as debt increases

TCJA Macro variables, closed economy



TCJA Fiscal variables, closed economy



TCJA Dynamic Analysis

DeBacker, Jason and Anderson Frailey, “[Revenue and Macroeconomic Effects of a 70% Marginal Tax Rate](#),” *Quantitative Notes*, 2019-1, (March 4, 2019).

- Can raise between \$5B and \$250B per year
- Reduce GDP in short run between 0.1% and 1.7%

70 Percent Federal Revenues Change

Table 1. Partial Equilibrium Revenue Estimates, Current Law vs. 70% Top MTRs

	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2019-2023	2019-2028
Current Law	2,834	2,968	3,089	3,214	3,349	3,495	3,650	4,039	4,219	4,409	15,455	35,267
70% top 2 brackets	3,042	3,179	3,305	3,436	3,578	3,735	3,902	4,170	4,358	4,554	16,540	37,259
Pct Change	7.32%	7.12%	6.98%	6.89%	6.85%	6.87%	6.91%	3.25%	3.28%	3.29%	7.02%	5.65%
70% top bracket	2,984	3,120	3,243	3,372	3,513	3,666	3,830	4,158	4,345	4,541	16,232	36,772
Pct Change	5.30%	5.12%	4.99%	4.91%	4.89%	4.89%	4.92%	2.95%	2.98%	2.99%	5.03%	4.27%
70% on \$1mm+	2,943	3,078	3,200	3,327	3,466	3,617	3,779	4,105	4,289	4,482	16,015	36,287
Pct Change	3.85%	3.70%	3.60%	3.52%	3.50%	3.50%	3.52%	1.65%	1.66%	1.66%	3.63%	2.89%
70% on \$10mm+	2,847	2,981	3,102	3,227	3,362	3,509	3,665	4,043	4,224	4,414	15,520	35,374
Pct Change	0.47%	0.43%	0.41%	0.40%	0.40%	0.40%	0.41%	0.10%	0.11%	0.11%	0.42%	0.31%

Dollar values in billions.

Revenue totals are for the individual income tax and payroll taxes.

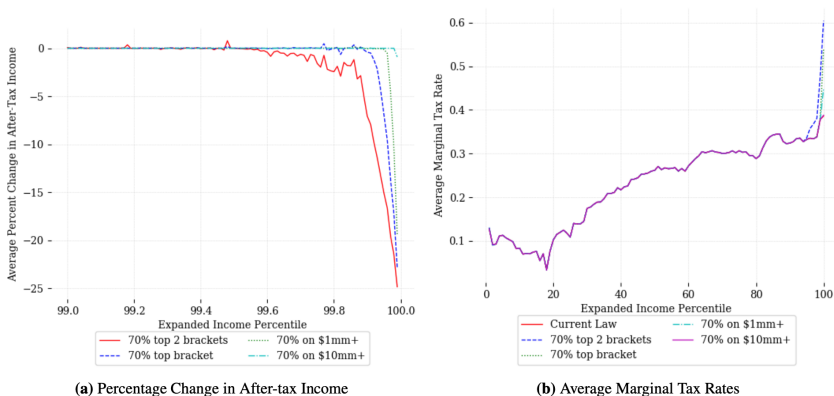
Percent changes are reported relative to current law.

Top two brackets and top bracket refer to the individual income tax brackets defined under current law.

Behavioral responses are found under an assumption of an elasticity of substitution of 0.25.

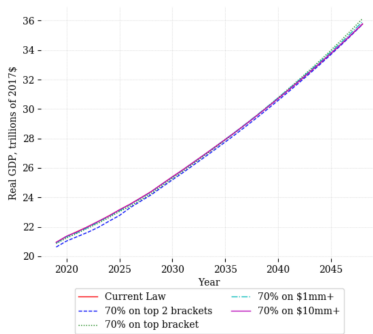
70 Percent Incidence

Figure 2. Distributional Impacts of a 70% Top Rate

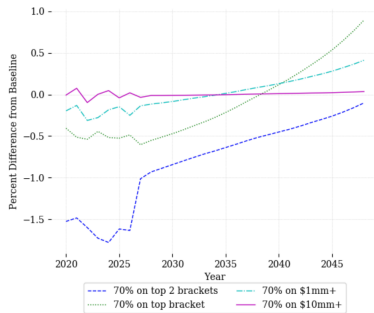


70 Percent GDP change

Figure 3. Real GDP, 2019-2049



(a) Gross Domestic Product



(b) Percentage Differences from Baseline GDP

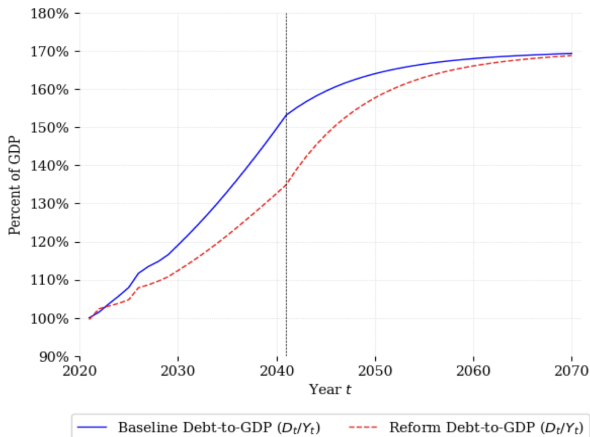
Analysis of President Biden's Campaign Proposals

DeBacker, Jason, Richard Evans, and Benjamin Page, “[A Detailed Macroeconomic Analysis of President Biden's 2020 Campaign Tax Proposals](#),” *Tax Policy Center Working Paper*, July, 2021.

- Debt reduction crowds in private capital in medium term
- We provide a sensitivity analysis to various parameter values

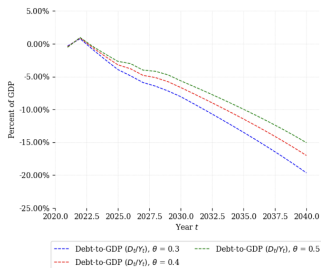
Biden Dynamic Analysis

Figure 6: Debt-to-GDP Ratio, Current Law Baseline vs. Biden Proposals

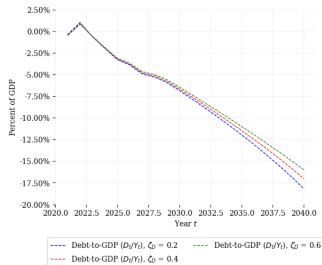


Biden Dynamic Analysis

Figure 6: Sensitivity of Differences in the Debt to GDP Ratio (Biden proposals - Current law baseline)



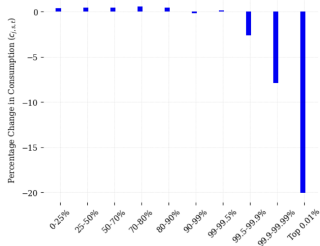
(a) Frisch



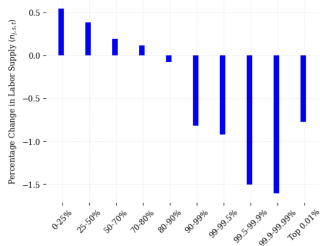
(b) Foreign Debt Purchases

Biden Dynamic Analysis

Figure 8: Behavioral Responses by Lifetime Income Group, 2021-2031



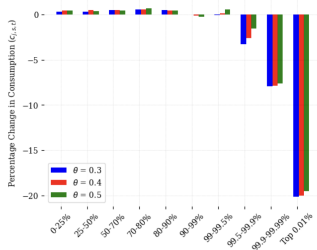
(a) Consumption



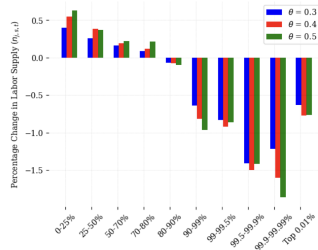
(b) Labor Supply

Biden Dynamic Analysis

Figure 1: Behavioral Responses by Lifetime Income Group, 2021-2031, Frisch Elasticity



(a) Consumption



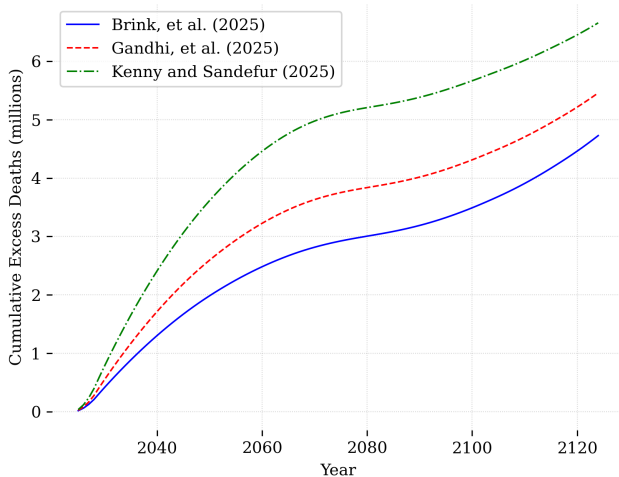
(b) Labor Supply

The Macroeconomics of Public Health Spending

DeBacker, Jason, Richard Evans, and Marcelo LaFluer, “The Economic Costs of Public Health Spending Cuts: HIV/AIDS and Tuberculosis in South Africa”, *Working Paper*, December, 2025.

- Withdrawal of US aid to South Africa’s HIV/TB programs leads to large economic costs

Forecasted Excess Deaths



GDP Effects of US Aid Cuts

Net Present Value of GDP Loss (billions of 2025 USD)

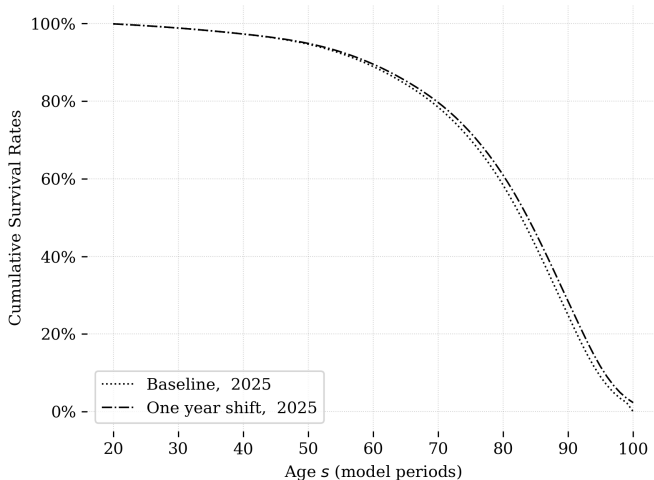
Discount rate	Low scenario Brink (2025)	Medium scenario Gandhi (2025)	High scenario KS (2025)
1%	-447.83	-2,170.70	-5,017.78
2%	-525.35	-1,428.99	-2,917.75
3%	-472.10	-979.31	-1,811.36
4%	-391.20	-696.53	-1,194.60
6%	-253.34	-387.97	-604.44

The Macroeconomics of Improving Biological Aging

DeBacker, Jason, Richard Evans, and Raiany Romanni, “The Macroeconomics of Improving Biological Aging”, *Working Paper*, January, 2026.

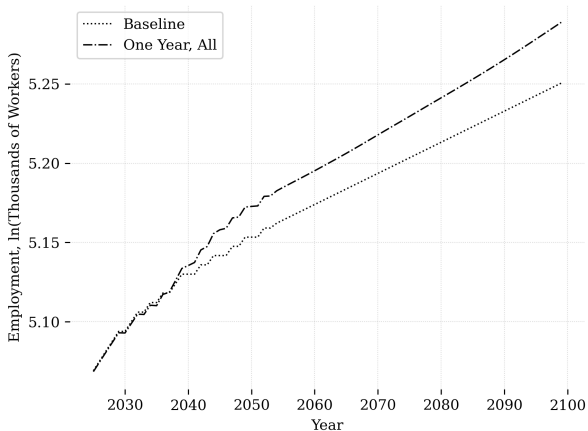
- Improving health outcomes leads to large macroeconomic benefits
- The time horizons over which health impacts affect the economy can be many decades

Changing Mortality



Employment Effects

Employment Paths, One Year Shift in Biological Age



GDP Effects

NPV of Change in GDP in Trillions of Constant 2025\$, One Year Shift in Biological Age

Discount Rate	Mortality	Productivity	Fertility	All
2%	98.52	35.75	73.21	209.86
4%	26.47	10.32	13.79	51.20
6%	8.63	3.69	2.77	15.29
Model r	10.40	4.47	3.25	18.06

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